SOEN 343: Software Architecture and Design
Project Phase I
Dishes Be Crazy

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I - Problem Definition

a. Problem Statement

The problem is	Currently, the barrier to entry into the restaurant space is the startup costs. Between paying rent, ordering industrial-level tools, and hiring staff, the initial costs can scale to more than 300 000$ (Ownr, 2026, "Calculating Restaurant Startup Costs" section).
Affects	Small businesses and individuals attempting to either get started in the food business, or earn a bit of money on the side. As such, many of these catering businesses rely on things such as facebook marketplace to manage orders.
The impact of which is	Small businesses struggle to advertise their food, provide service to customers, and grow their business. These costs sway ambitious home cooks away from the chance of creating their own business and sharing their meals with the community.
A successful solution would be	Providing an easy-to-use hub to market and share the food of home cooks with the community. This way, users can put less time, money, and stress into the logistical side of the food industry, and more time doing the things they love. Additionally, it provides a platform for users to support local businesses and give back to the community.

b. Product Position Statement

For	Homecooks, couriers and members of the community
Who	Are looking for an intuitive, local, and easy-to-use food sharing system to manage and support small businesses
Dishes Be Crazy	is a home-cooked meal delivery service
That	allows homecooks to advertise and distribute their food without having to own their own restaurant space.
Unlike	Uber Eats, Doordash, Facebook marketplace and Skip the Dishes

Our product	Is catered towards individuals and communities rather than corporations and businesses with established retail spaces, providing more freedom and support to businesses than other alternatives.

II - Product Overview

a. Product Perspective

Dishes Be Crazy is a standalone product. We considered positioning it as an ordering and delivery system for an existing restaurant or franchise, which would require integrating with that business's point-of-sale, menu and inventory systems. We rejected this because our target users are independent home cooks who have no existing systems to integrate with; for them, the platform itself is the storefront, order manager and delivery coordinator. It does not replace an existing system: home cooks today typically rely on informal channels such as Facebook Marketplace to sell and coordinate orders.

What differentiates Dishes Be Crazy from traditional delivery apps is the social layer. Rather than a static directory of restaurants, customers can discover food through a feed of posts that cooks publish about the dishes they are currently offering. Cooks gain visibility and a following through their content rather than through paid placement or an established storefront. In essence, this allows a new home cook with a single good dish to reach customers the same way an established one does which is central to our goal: lowering the barrier of entry into the food business.

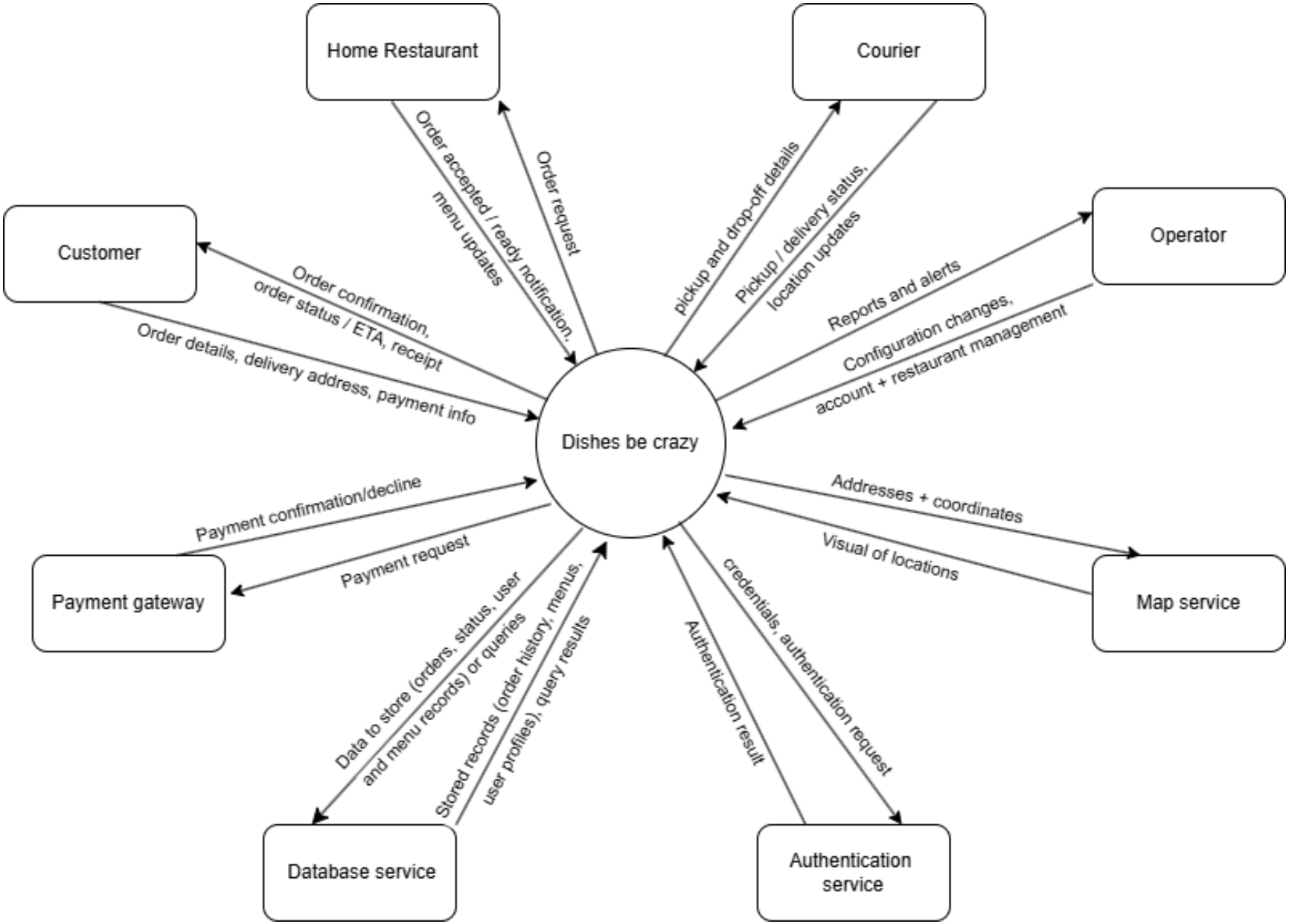
b. Assumptions and dependencies

Assumptions	Dependencies
The customers and the restaurants will have a valid home address	Google Maps API will be the core of how we enable location tracking
All customers will have valid forms of digital payment to pay for their orders	Our application will use Stripe for payment authorization and validation
Restaurant staff will post accurate business hours, menus and dish availabilities	Supabase will be used to store saved data tables and records for the application
A platform operator is always on call to deal with exceptional circumstances or system management	Clerk will handle user authentication and ensure secure access to the application
Every cook selling through the platform holds the food-handling and food-sale permits required where they operate (in Quebec, issued by the MAPAQ) and complies with applicable food-safety rules.	Provincial food regulations. If home kitchens can't be permitted, or the rules change, the product must change, e.g. restrict cooks to licensed shared kitchens or add permit verification at sign-up.

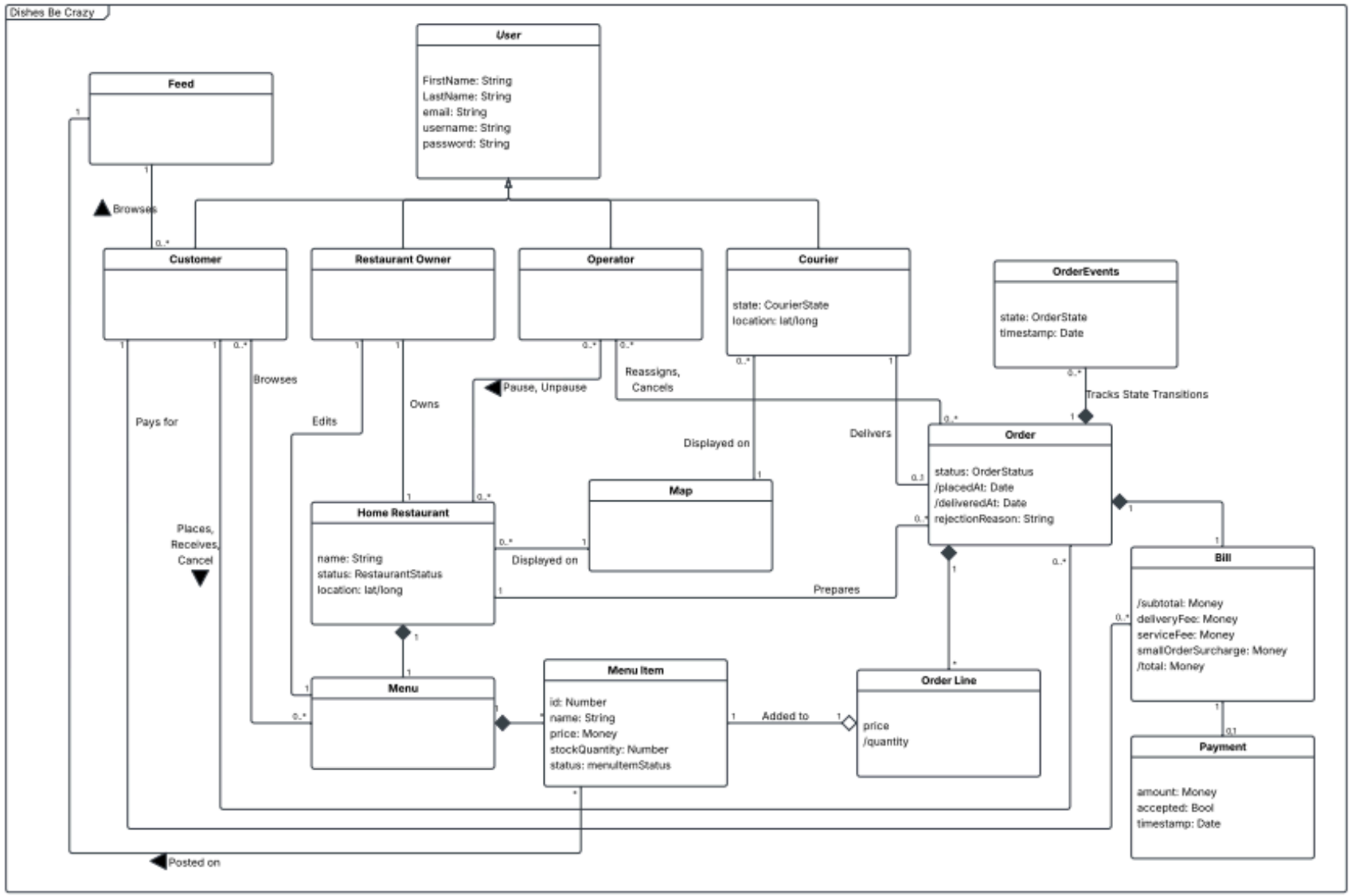
III - Technology planned:

a. **Control version system:**
   GitHub
b. **Team collaboration**
   Discord
c. **Design and modelling work**
   Lucid.app, VisualParadigm
d. **Development framework**
   React with typescript (frontend) Java Springboot (backend)
e. **Coding**
   TypeScript, Java, HTML/CSS, SQL
f. **GenAI tools used**
   Claude/Claude Code, ChatGPT/Codex

IV - Context diagram



V - Domain model



Domain Model: **Mosby Boys - Domain Diagram - SOEN 343_ Lucidchart.pdf**

VI - Reference

This work was primarily human-created. AI was used to make stylistic edits, such as changes to wording, and clarity. AI-generated content was reviewed and approved. The following model(s) or application(s) were used: Claude Opus 5.5. ← right here**

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Team, O. (2026, April 7). *Advice from restohub: Restaurant startup costs*. Ownr Blog. https://www.ownr.co/blog/restohub-restaurant-startup-costs/